

The Dark Side of Docking Scores

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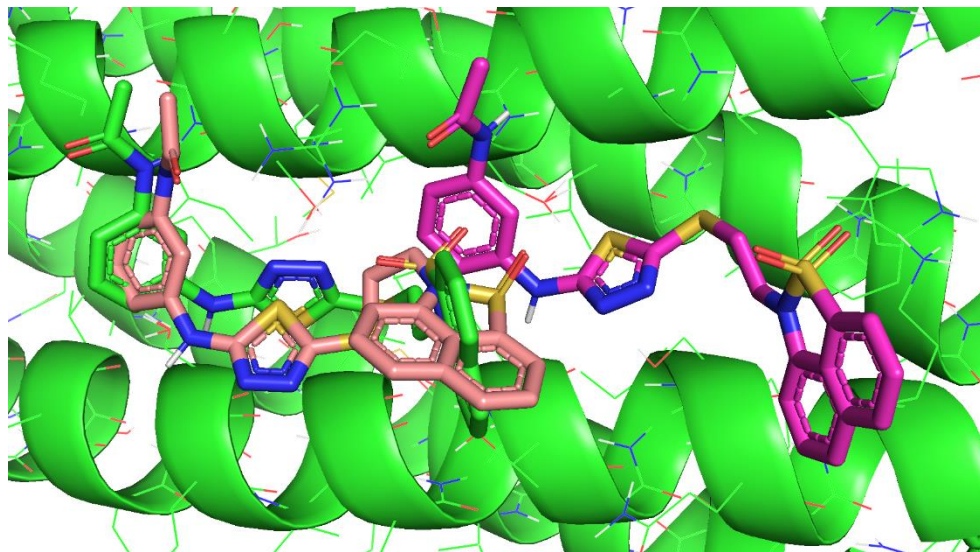
Outline

- ▶ The Docking/Scoring Problem...
 - ▶ Ugly things behind the scenes... it's hard to be rigorous!
- ▶ What is a docking benchmark worth?
 - ▶ Sometimes it “works” – but what does that mean?
- ▶ All models are wrong, but some are useful!

Beware! Docking begins way before the actual “docking” – with the preparation of the protein site, and standardization of ligand structures

If you did not correctly specify the protonation states of your ligand molecules, or you have forgotten to delete the large detergent molecule X-rayed in the protein active site... don't blame the docking software for your epic failure!

Docking of a Ligand Results in an *Ensemble of Poses*, which are characterized:



- ▶ By their Energy E
 - ▶ More stable poses are *a priori* more relevant!
- ▶ Alternatively, by a Contact Fingerprint
 - ▶ Poses establishing key contacts already seen in strong binders are more relevant!

Nice, but... we are not really interested in the detailed biography of each of these poses!

What docking is all about is predict how active the ligand will be – somehow based on the properties of those poses!

Docking-driven Virtual Screening...

¡Viva la *Energia Libre*,
camaradas!

$$F_{dock} = -k_B T \ln \sum_{docked} \exp\left(-\frac{E_i}{k_B T}\right)$$

$$F_{unbound} = -k_B T \ln \sum_{unbound} \exp\left(-\frac{E_i}{k_B T}\right)$$

$$\Delta F = F_{docked} - F_{unbound}$$

ΔE ?



Scoring Functions: The Revenge of Empiricism over Boltzmann's Ensemble Physics

- ▶ We can **not** apply rigorous statistical physics:
 - ▶ We cannot – *at least not without using large-scale computing facilities* – enumerate all the relevant states of the Ensemble.
 - ▶ The inaccuracy of force field energies goes way beyond $k_B T \approx 0.6$ kcal/mol – so much for $\exp(-E/k_B T)$
- ▶ The state-of-the-art response was to *fit* a QSPR equation, aiming to predict the binding ΔF from contacts seen in the most stable pose returned by docking, or in crystal structures...
 - ▶ Such a construct is called a **scoring function**.

$$\Delta F = \alpha \times \text{ContactCount}_{HBond}^{site-lig} + \beta \times \text{ContactCount}_{Hphobic}^{site-lig} + \gamma \times E_{vdW}^{site-lig} + \delta \times \text{BlockedTorsionCount} + \dots$$



BUT...

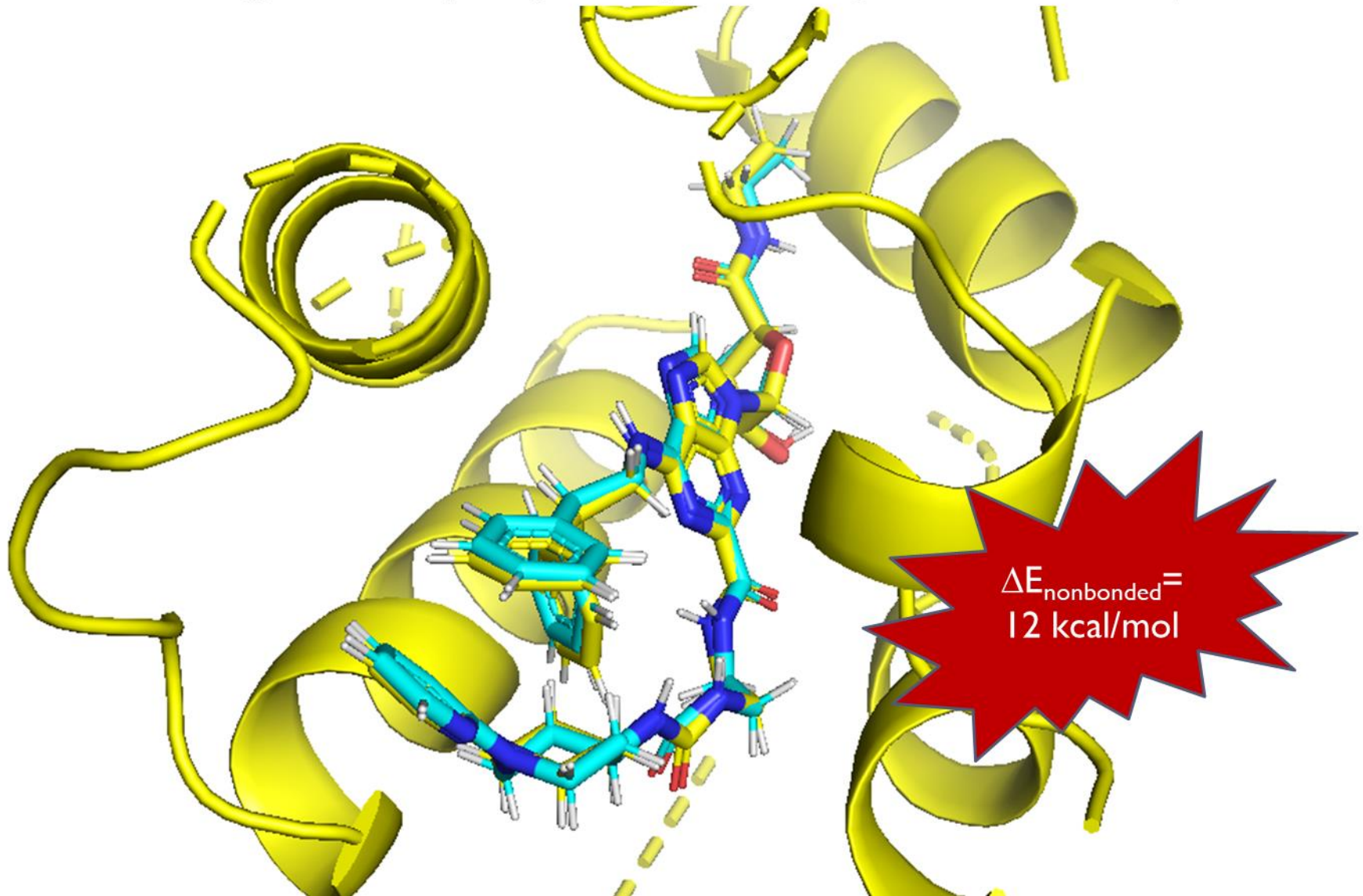
... let's sit back and think about what all this implies!

1. Type Mismatch Error ?

- ▶ The Free Energy is the property of an entire *conformational space zone*, hosting an *ensemble* of (very) many possible geometries...
- ▶ Yet, the scoring function approach claims to predict it based on a *single « representative » geometry*.
- ▶ ... not to mention that ΔF calculation also requires a sampling of the ensemble of unbound ligand conformers – **almost always ignored in commercial docking tools**.

2. Fine, but... which « representative » geometry ?

PDB versus Energy-minimized pose (S4MPLE) of 3QAK ligand in the A2A receptor site



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Two Classes of Docking Benchmarks

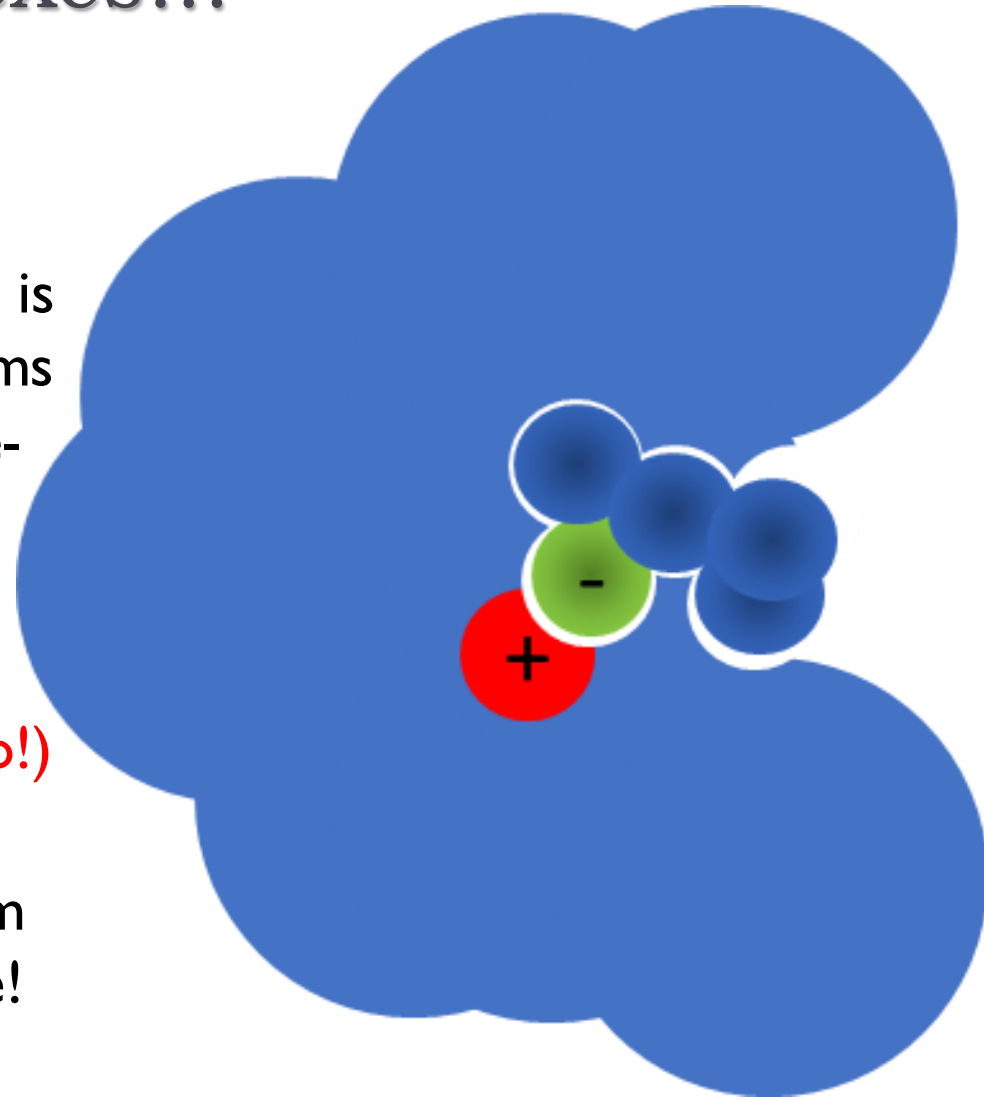
- ▶ **Pose Benchmarking (Pbench)**: is the tool able to predict the X-ray pose of known ligands?
 - ▶ ... starting with the protein site geometry as seen in that actual X-ray structure, in presence of the concerned ligand (“Redocking”)
 - ▶ ... starting with a protein site geometry X-rayed in presence of a *different* ligand (“Cross-docking”)
- ▶ **Affinity Benchmarking (Abench)**: dock and score a series of known ligands, then check whether the ranking by docking score of **best** pose matches actual ranking by affinity
 - ▶ Subtlety: if the best pose is not the native one, one should ignore that ligand in Abench...
 - ▶ but in practice that pose is in many cases not even known! So what if we get great Abench performance... based on wrong poses?

Pose Benchmarking (Pbench) is most popular!

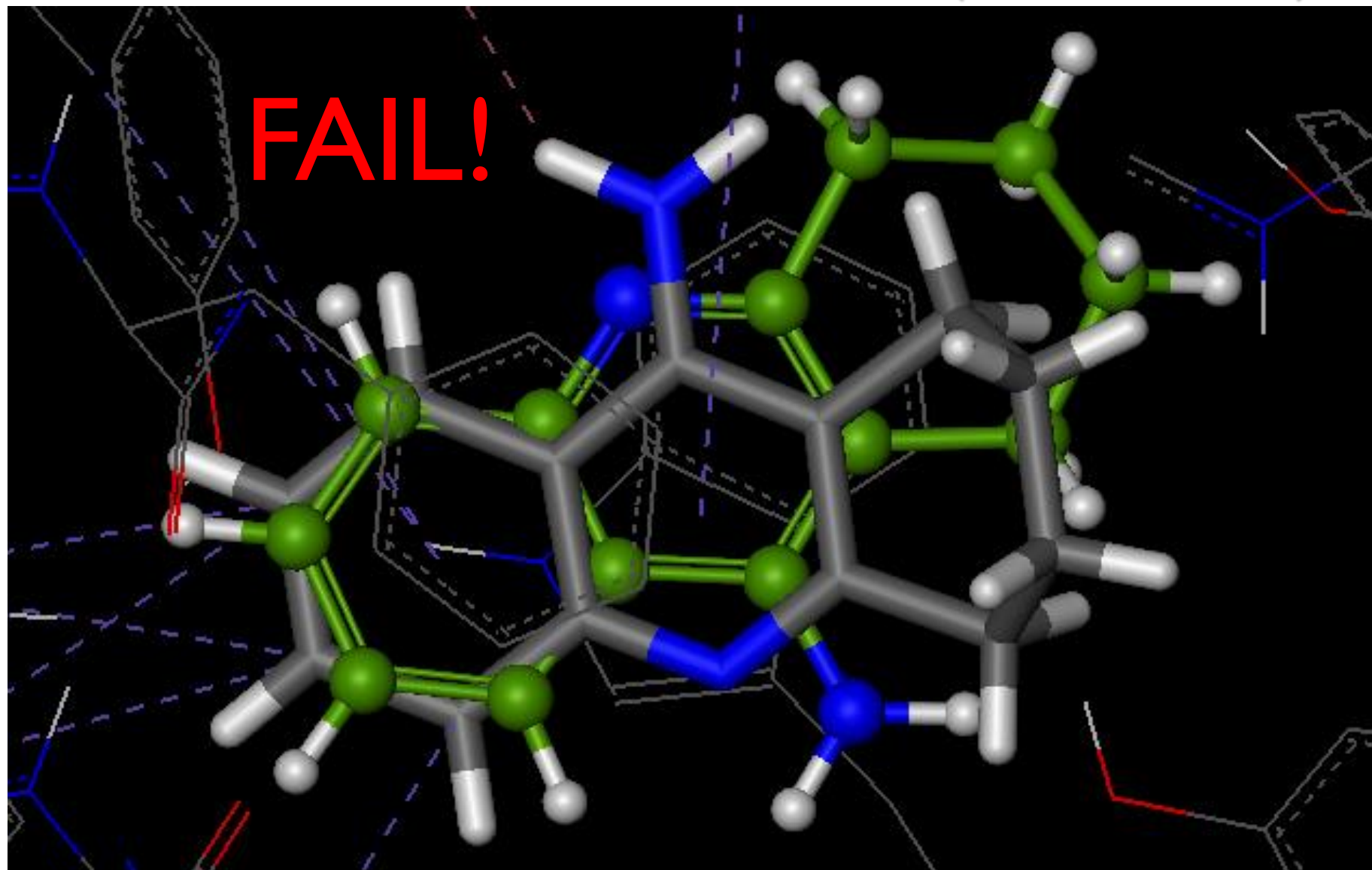
- ▶ Requires, for each ligand-target complex LT , the computed poses LT^p , to be **ranked** by their stability/score, and compared to the **native** geometry LT^0
- ▶ $P^k(\tau) = \frac{1}{N_{LT}} \sum_{i=1}^{N_{LT}} \delta(\exists p \in [1, k] \text{ so that } \Delta[LT^p, LT^0] \leq \tau)$
- ▶ Checks, for each complex, if amongst the best k poses, there are some less dissimilar to the native (according to some metric Δ – typically RMSD) than threshold τ (typically 2 Å), where $\delta(\text{statement})$ is a Boolean-to-Bit convertor.
- ▶ Strict Pbench at $k=1$ is the most “honest” one – because in docking one rarely looks at poses other than the best-ranked.
- ▶ Other metrics – like contact FP dissimilarity – may be more meaningful than RMSD... but are almost never used!
- ▶ May concern a diverse panel of protein targets... or only the protein of interest

An example, using the Astex Diverse Set of protein-ligand complexes...

- ▶ **Docking works!**
 - ▶ At least when the benchmarking is conciliant! Most docking programs manage to generate some native-like pose within their outputs...
 - ▶ **However... many of those examples are way too obvious (will be correct whatever you do!)**
 - ▶ ... while others are too difficult (*water-mediated?*), so getting them right may only happen by chance!

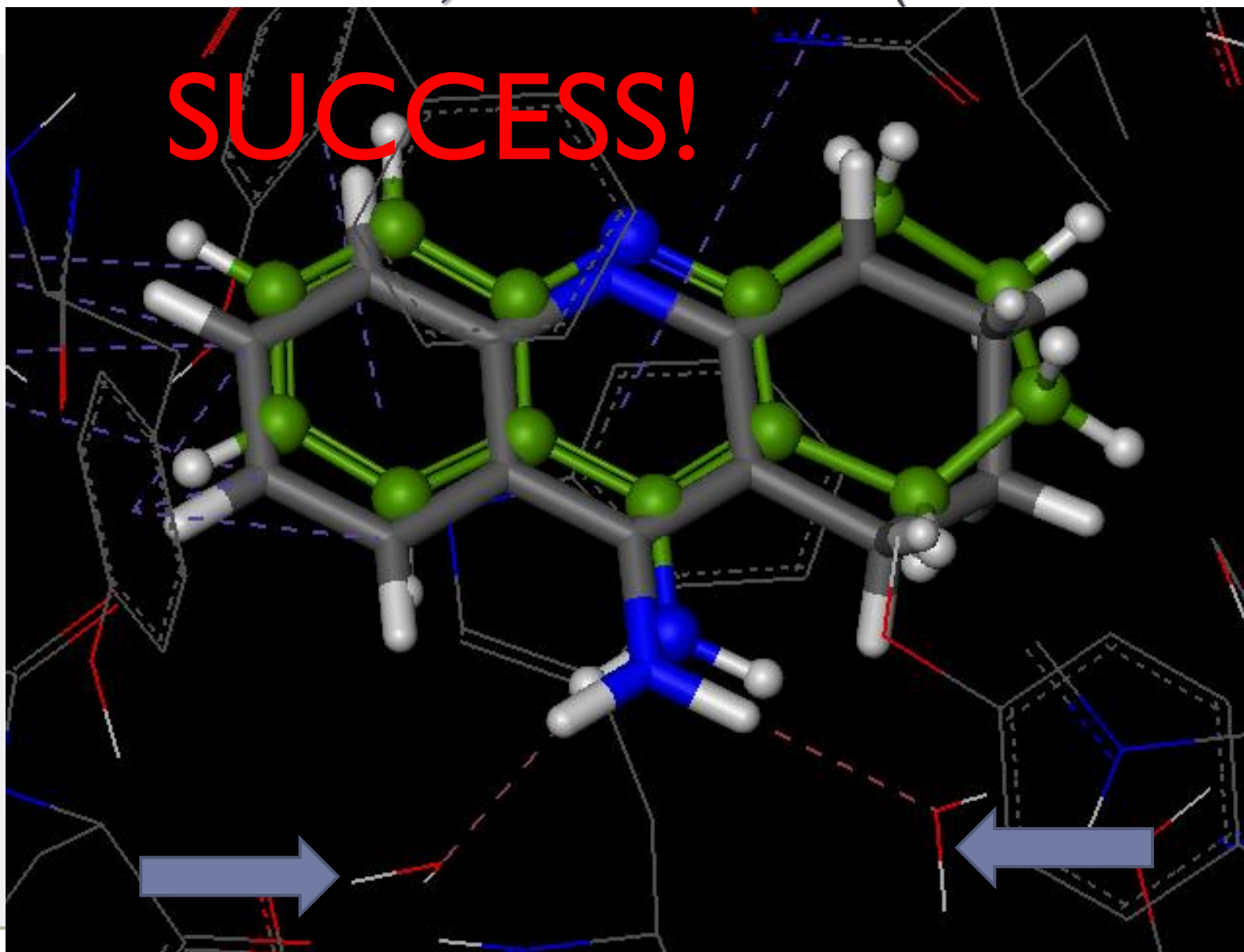


Case Study: Redocking Tacrine into Acetyl Choline Esterase, with LeadIT (BioSolveIT)

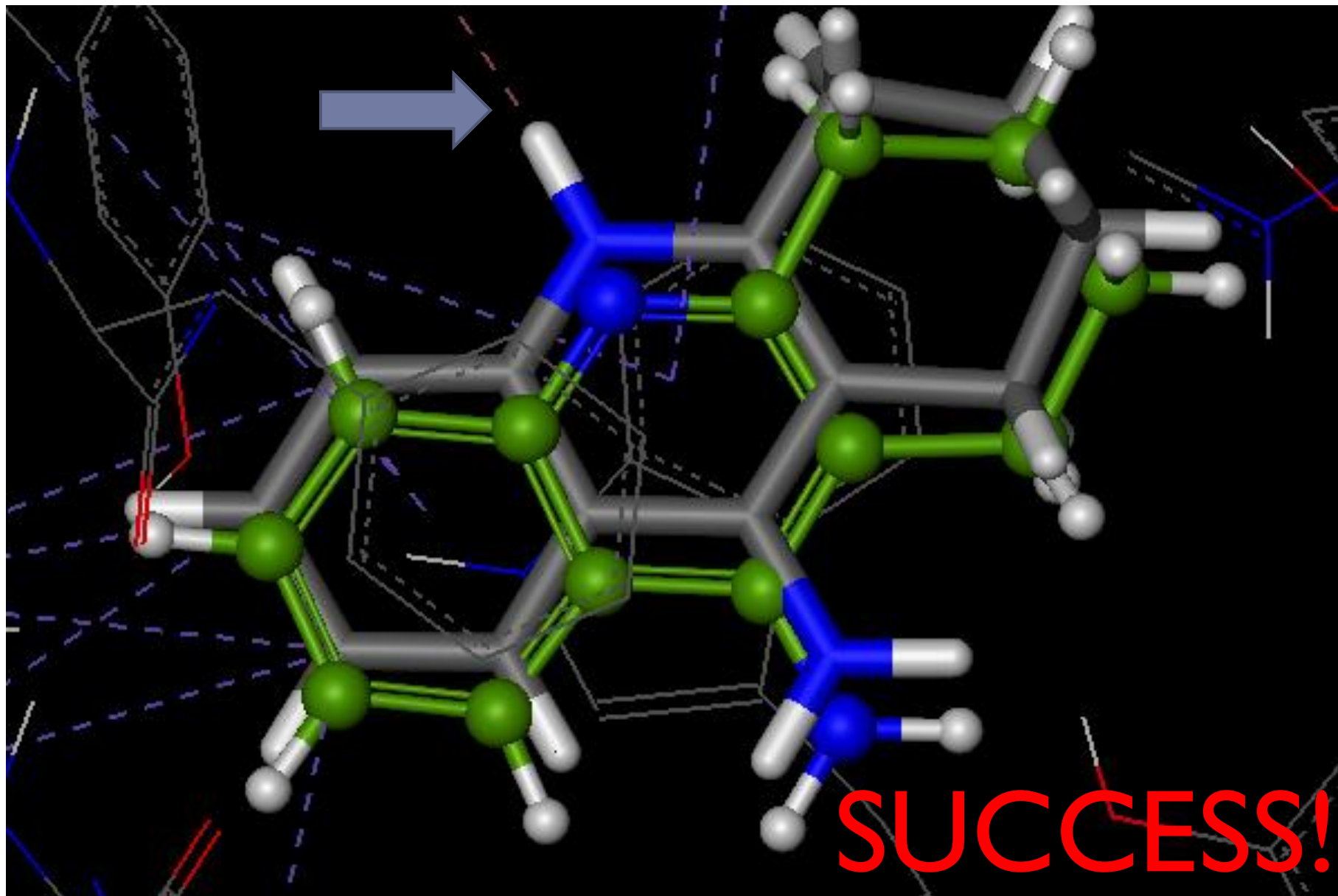


Case Study: Redocking Tacrine into Acetyl Choline Esterase, with LeadIT (BioSolveIT)

▶ **SUCCESS!**



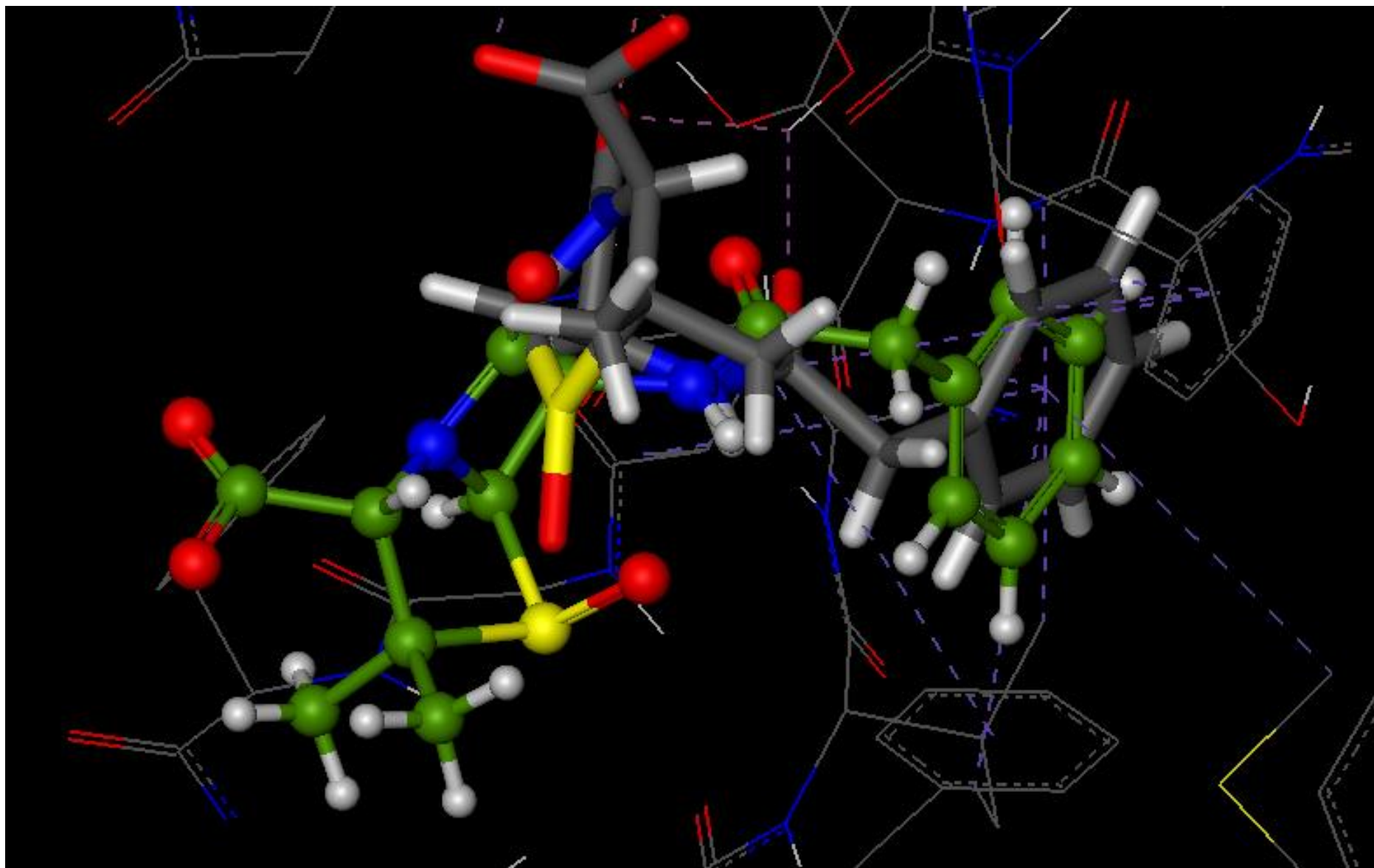
Protonate as LeadIt thinks? Not really...



Right for the Wrong Reasons...

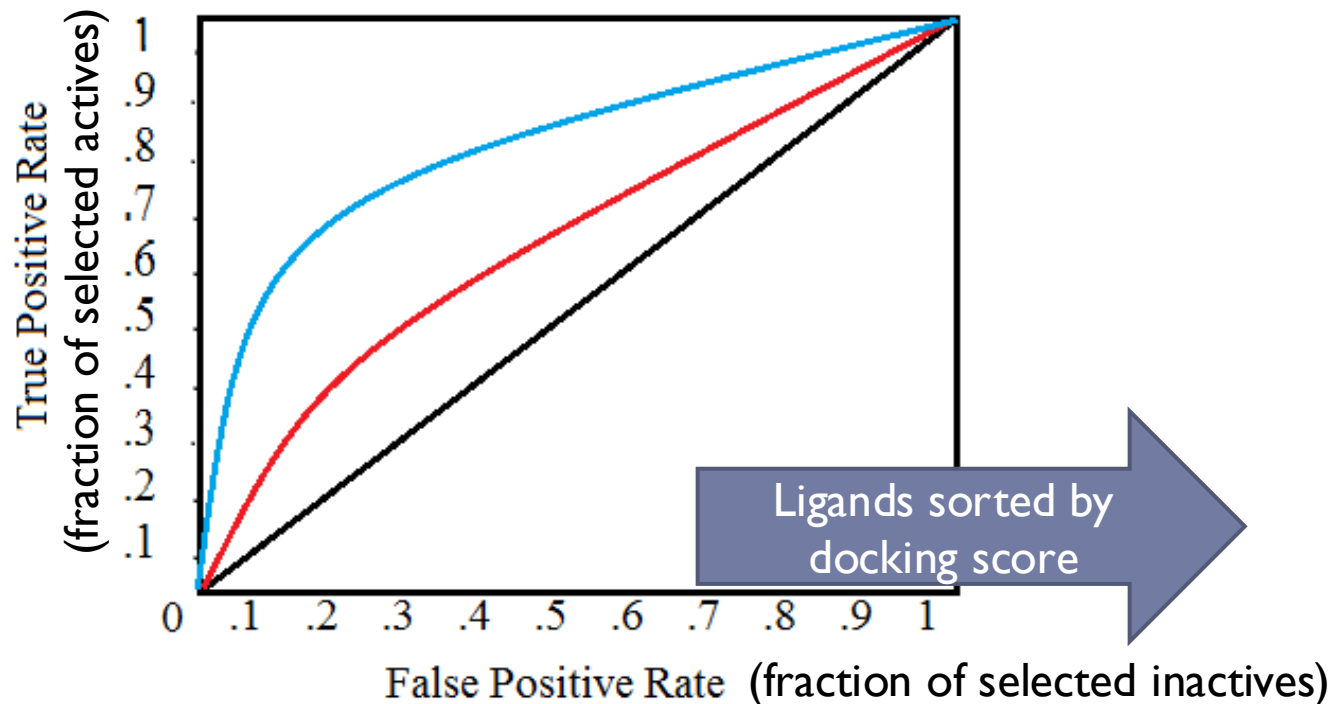
- ▶ Correct protonation *with* crystallographic waters also leads to the correct docking pose...
- ▶ However, capturing a single of the two key mechanisms defining the orientation of Tacrine turns out to be « good enough » to pass the benchmark
- ▶ Sometimes, docking tools are good... and sometimes they are lucky
 - ▶ Getting the correct protonation state of the ligand in its *unbound* state is difficult enough – best done by the user at ligand preparation stage...
 - ▶ But even a perfect acidity constant predictor may fail if protonation states change *during* the docking!

Pose benchmark failure? Maybe not!



Affinity Benchmarking (Abench): Good Docking = Nice ROC Curves!

- ▶ Successful docking should systematically return better affinity values for actual actives, compared to non-binders



So – does success in Pose Benchmarking imply
success in Affinity Benchmarking, and
reciprocally?

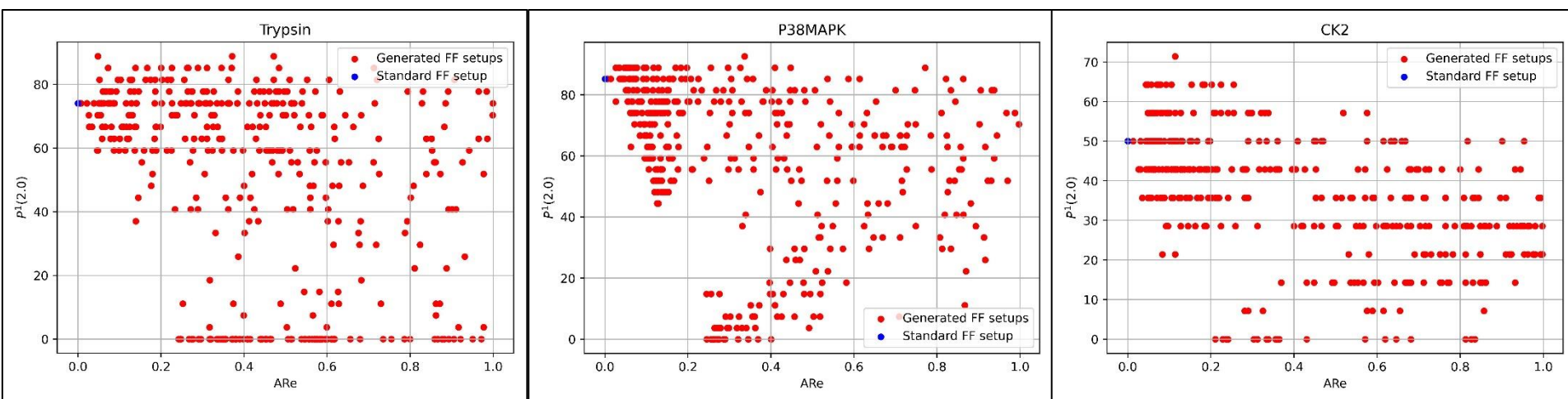
NO WAY!

A Simple Question: What is the Impact of Changing an Empirical Parameter Value of my Scoring Function?

- ▶ Like: will changing the relative dielectric constant from 2.0 to 3.5 have a measurable influence on my docking calculations?
- ▶ If the relative order of score-ranked poses, or the relative order of score-ranked ligands of a target does not change a lot, the impact is likely low!
- ▶ Relate the impact to the Spearman Rank correlation coefficients ρ of lists obtained with the new setup, compared to lists at “native” setup.
- ▶ Pose Reordering: $PRe(\vec{p}_1, \vec{p}_2) = 1 - \langle \rho_{pose\ list(LT)}(\vec{p}_1, \vec{p}_2) \rangle_{LT}$
- ▶ Affinity Reordering: $ARE(\vec{p}_1, \vec{p}_2) = 1 - \rho_{ligand\ list}(\vec{p}_1, \vec{p}_2)$
- ▶ where the two compared parameter configurations are depicted as vectors $\vec{p}_1, \vec{p}_2$.
- ▶ PRe is taken as the average degree of reordering of poses over all targets, ARE looks at ligands (all targets confounded) ranked by the scores of their best pose.

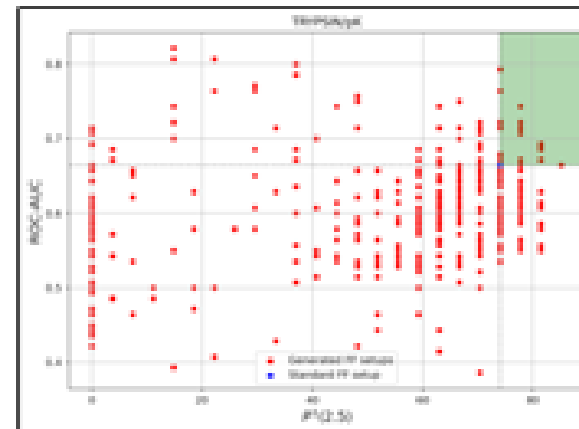
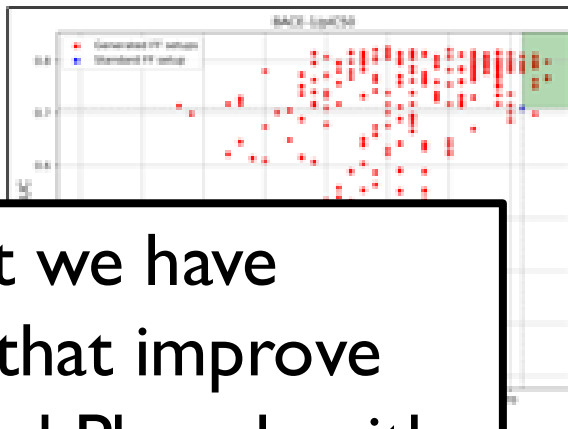
Optimizing Pbench criteria may NOT simultaneously grant maximal Abench success!

- ▶ A “*standard*” docking parameter set* for the S4MPLE tool (blue dot) guarantees good Pbench statistics... but many other setups (out of 400 random parameterization schemes) lead to *similar or better* Pbench success, **all while proposing significantly differing ligand rank orders!**



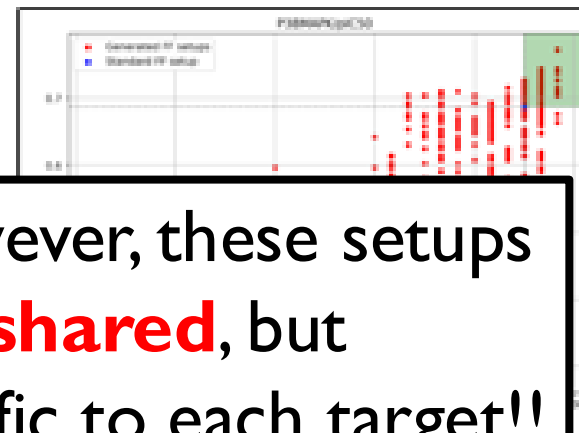
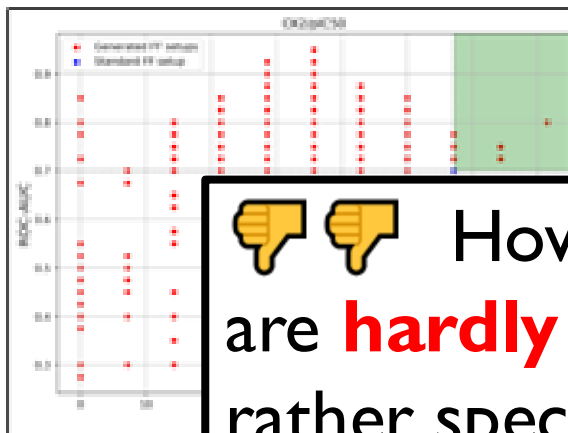
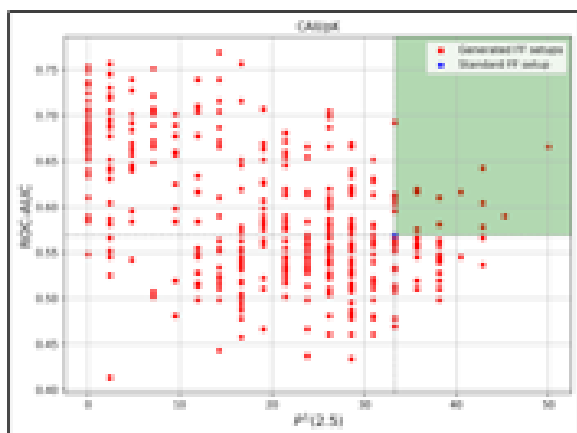
* Hoffer, L. et. al. S4MPLE-Sampler for Multiple Protein-Ligand Entities: Methodology and Rigid-Site Docking Benchmarking. *Molecules*, 20 (5), 8997-9028 (2015)

Pbench and Abench success are not strongly correlated
– even when looking at ligand series of a same target!



👍 Ok, at least we have **some setups** that improve both Abench and Pbench with respect to the standard setup!

Trypsin



👎👎 However, these setups are **hardly shared**, but rather specific to each target!!

CAII.

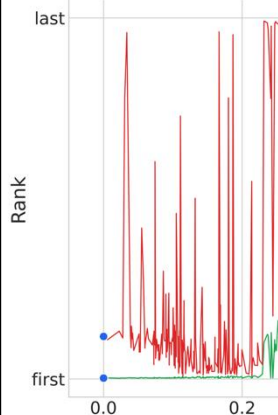
CK2

P38MAPK

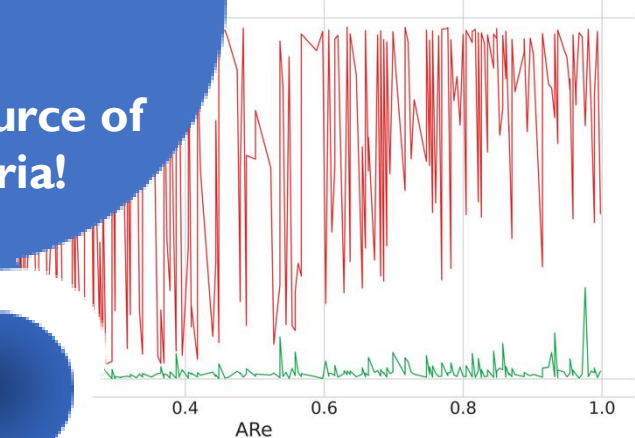
As Expected, Some Ligands are Highly Sensitive to Docking Parameters

Our cliché “infallible ligand” is unfortunately not a far-fetched Gedankenexperiment, but a source of much inflated benchmark criteria!

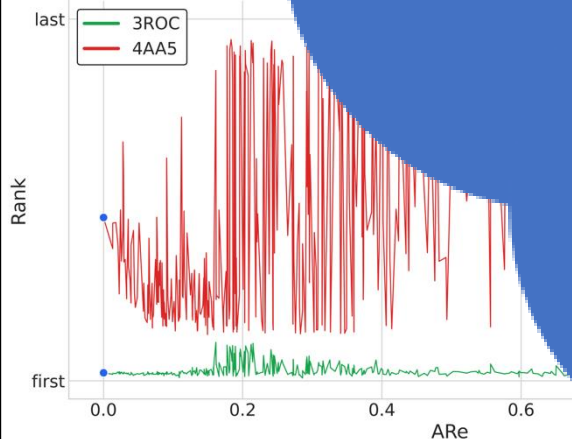
BACE-1



CK2



Rank
last
first



The native-like pose may be very little with respect to docking scheme.

As not fluctuate, the consistently number of compounds to bind.

pick your score from compound sets! An outlier in your set will give you great docking scores!

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So, what do we learn from this?

- ▶ If there is one real active among the 10 top-scored ligands, the docking tool achieved its goal...
 - ▶ ... because, without docking, 100+ compounds would have to be tested in blind High Throughput Screening in order to find one active.
- ▶ Nonetheless, docking is more often wrong than right – therefore, it makes no sense to run docking if you do not intend to experimentally validate the selected compounds
 - ▶ Docking does not provide *answers*, but generates useful *working hypotheses* for drug design!
 - ▶ It is certainly *not* the answer to your quest to understand an action mechanism!
- ▶ Protonation states, water mediation, protein flexibility, covalent binding are still waiting for solutions
 - ▶ ... at least for solutions fast enough for virtual screening.

So, what do we learn from this?

- ▶ Predicting the correct pose does not imply that the underlying mathematical model is correct...
- ▶ ... and does not imply that Affinity Ranking will be correct either
- ▶ Tuning a Scoring Function (*if you can close an eye on all the fundamental incoherencies such a construct implies*) may only succeed if a maximum of experimental data (both Pbench AND Abench challenges) are included,
 - ▶ at huge computational cost...
- ▶ And within a well-defined Applicability Domain!
 - ▶ ... or, nobody uses this classical QSAR concept in conjunction to scoring functions, albeit interaction fingerprint matching implicitly plays this role!
- ▶ Take your time! Dock fewer candidates, but well, rather than billions with bogus scores...

Like it or not, Docking has a Key Role in Modern Drug Discovery!

- ▶ Thank you for listening...
and don't mess with the
dockers!

Forbes

FORBES > BUSINESS > MANUFACTURING

Why The Dock Workers' Strike Should Be A Cautionary Tale For Every Manufacturer

Ethan Karp Contributor 
I write about transforming businesses through technology & innovation.







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